

AIG130 – Lab 1

Objective

This lab focuses on understanding the role of cloud computing in machine learning, exploring popular cloud platforms, and discussing the challenges and considerations in implementing cloud solutions for machine learning projects.

Group Formation

Students can work individually or form groups of up to 4 members. Collaboration and effective communication are encouraged for those working in groups.

Tasks and Instructions

Why Cloud Computing for Machine Learning?

- **Objective:** Investigate and analyze the benefits and rationale for using cloud computing in machine learning.
- **Task:**
 - ⇒ Research and compile information on how cloud computing supports machine learning projects in terms of scalability, cost, accessibility, and computational power.
 - ⇒ Prepare a report section highlighting case studies or examples of successful machine learning projects that leveraged cloud computing.

Popular Cloud Platforms for Machine Learning

- **Objective:** Compare and contrast different cloud platforms and their offerings for machine learning.
- **Task:**
 - ⇒ Select at least two popular cloud platforms (e.g., AWS, Google Cloud Platform, Microsoft Azure).
 - ⇒ Evaluate their machine learning services, ease of use, scalability, and pricing.
 - ⇒ Create a comparative analysis section in the report, including tables or charts for visual comparison.

Challenges and Considerations

- **Objective:** Identify and discuss the challenges and considerations in using cloud computing for machine learning.
- **Task:**
 - ⇒ Research issues such as data security, compliance, cost management, and performance optimization.
 - ⇒ Propose solutions or best practices to address these challenges.
 - ⇒ Include a section in your report discussing these challenges and potential strategies to overcome them.

Deliverables

Report: Submit a PDF report covering all three tasks. Include:

- An introduction and conclusion.
- Sections for each task with headings.
- Visual elements (charts, tables) where applicable.

Presentation: Prepare and deliver a presentation of your findings in class. The presentation is also required to be submitted in Blackboard. It can be as PDF, PPTX, Link, etc.

- Duration: Up to 15 minutes per group.
- Cover key points and findings from the report.
- Encourage questions and discussions.

Individual Reflection

Each student must submit a separate reflection document answering the following:

1. If you worked individually, why did you choose to go solo? What challenges did you encounter?
2. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?

Submission

Submissions are done through the provided links in Blackboard. There are three links:

1. Group report (one submission per group): 4%
2. Individual reflection: 2%
3. Presentation file/link: 2%

Assessment Criteria

- Quality and depth of research and analysis.
- Clarity and organization of the report.
- Effectiveness of the presentation.
- Thoughtfulness of the individual reflection.